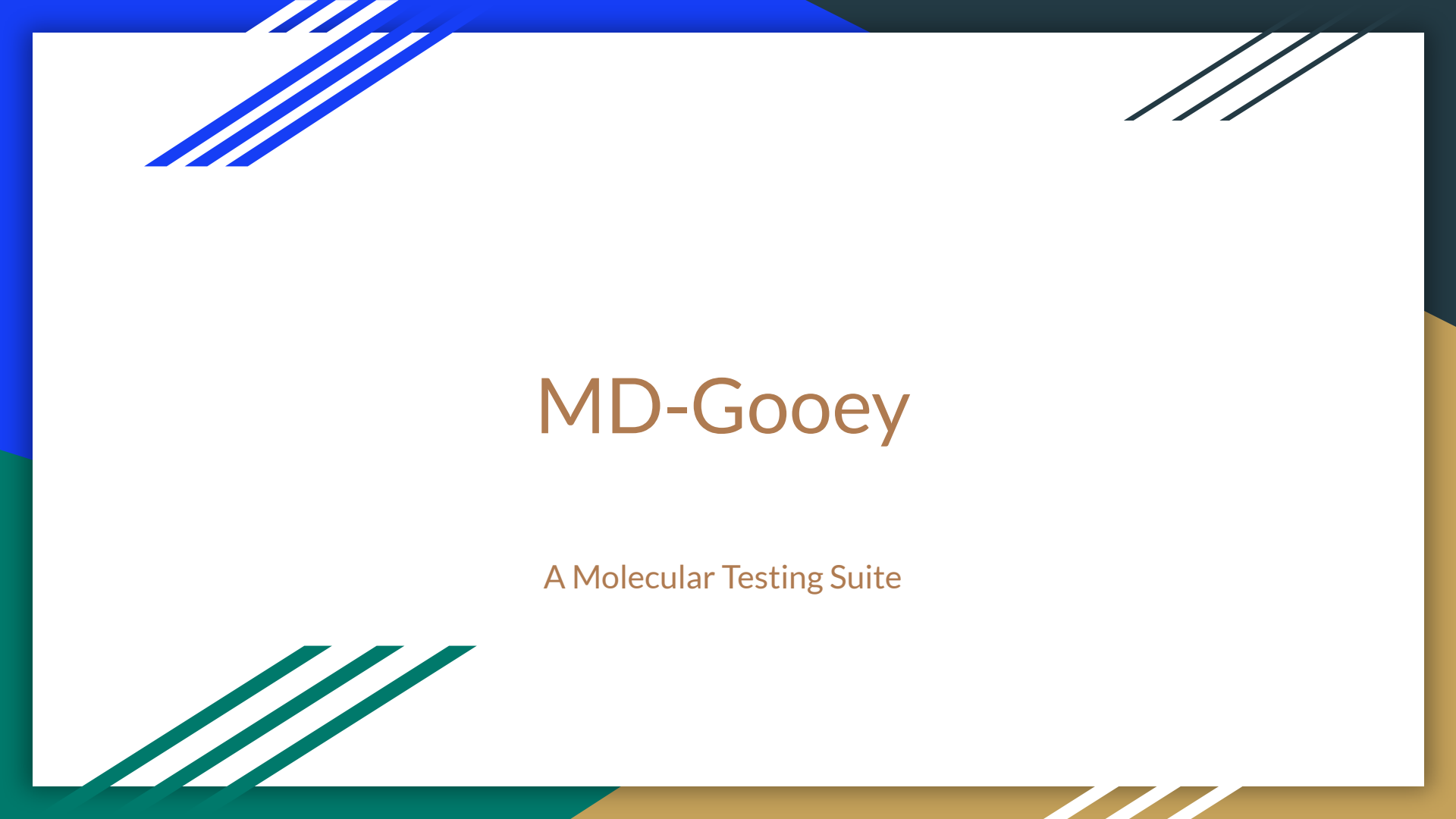




MD-Gooey

A Molecular Testing Suite

Our Customer

Her work

- **More broadly:** Mechanical Properties of Biological Macromolecules
- **What we're helping with:** Improving functionality of the site she designed, and Lauren Campbell developed during her research with Dr. Dionisio

Consultant/Manager- Dr. Dionisio

Dr. Zara Alavi



-
- (a) new_processed.gro**
- | Atom | x | y | z |
|------|--------|---------|--------|
| 1 | 10.460 | -14.370 | 10.470 |
| 2 | 10.460 | -14.370 | 10.470 |
| 3 | 10.460 | -14.370 | 10.470 |
| 4 | 10.460 | -14.370 | 10.470 |
| 5 | 10.460 | -14.370 | 10.470 |
| 6 | 10.460 | -14.370 | 10.470 |
| 7 | 10.460 | -14.370 | 10.470 |
| 8 | 10.460 | -14.370 | 10.470 |
| 9 | 10.460 | -14.370 | 10.470 |
| 10 | 10.460 | -14.370 | 10.470 |
| 11 | 10.460 | -14.370 | 10.470 |
| 12 | 10.460 | -14.370 | 10.470 |
| 13 | 10.460 | -14.370 | 10.470 |
| 14 | 10.460 | -14.370 | 10.470 |
| 15 | 10.460 | -14.370 | 10.470 |
| 16 | 10.460 | -14.370 | 10.470 |
| 17 | 10.460 | -14.370 | 10.470 |
| 18 | 10.460 | -14.370 | 10.470 |
| 19 | 10.460 | -14.370 | 10.470 |
| 20 | 10.460 | -14.370 | 10.470 |
| 21 | 10.460 | -14.370 | 10.470 |
| 22 | 10.460 | -14.370 | 10.470 |
| 23 | 10.460 | -14.370 | 10.470 |
| 24 | 10.460 | -14.370 | 10.470 |
| 25 | 10.460 | -14.370 | 10.470 |
| 26 | 10.460 | -14.370 | 10.470 |
| 27 | 10.460 | -14.370 | 10.470 |
| 28 | 10.460 | -14.370 | 10.470 |
| 29 | 10.460 | -14.370 | 10.470 |
| 30 | 10.460 | -14.370 | 10.470 |
| 31 | 10.460 | -14.370 | 10.470 |
| 32 | 10.460 | -14.370 | 10.470 |
| 33 | 10.460 | -14.370 | 10.470 |
| 34 | 10.460 | -14.370 | 10.470 |
| 35 | 10.460 | -14.370 | 10.470 |
| 36 | 10.460 | -14.370 | 10.470 |
| 37 | 10.460 | -14.370 | 10.470 |
| 38 | 10.460 | -14.370 | 10.470 |
| 39 | 10.460 | -14.370 | 10.470 |
| 40 | 10.460 | -14.370 | 10.470 |
| 41 | 10.460 | -14.370 | 10.470 |
| 42 | 10.460 | -14.370 | 10.470 |
| 43 | 10.460 | -14.370 | 10.470 |
| 44 | 10.460 | -14.370 | 10.470 |
| 45 | 10.460 | -14.370 | 10.470 |
| 46 | 10.460 | -14.370 | 10.470 |
| 47 | 10.460 | -14.370 | 10.470 |
| 48 | 10.460 | -14.370 | 10.470 |
| 49 | 10.460 | -14.370 | 10.470 |
| 50 | 10.460 | -14.370 | 10.470 |
| 51 | 10.460 | -14.370 | 10.470 |
| 52 | 10.460 | -14.370 | 10.470 |
| 53 | 10.460 | -14.370 | 10.470 |
| 54 | 10.460 | -14.370 | 10.470 |
| 55 | 10.460 | -14.370 | 10.470 |
| 56 | 10.460 | -14.370 | 10.470 |
| 57 | 10.460 | -14.370 | 10.470 |
| 58 | 10.460 | -14.370 | 10.470 |
| 59 | 10.460 | -14.370 | 10.470 |
| 60 | 10.460 | -14.370 | 10.470 |
| 61 | 10.460 | -14.370 | 10.470 |
| 62 | 10.460 | -14.370 | 10.470 |
| 63 | 10.460 | -14.370 | 10.470 |
| 64 | 10.460 | -14.370 | 10.470 |
| 65 | 10.460 | -14.370 | 10.470 |
| 66 | 10.460 | -14.370 | 10.470 |
| 67 | 10.460 | -14.370 | 10.470 |
| 68 | 10.460 | -14.370 | 10.470 |
| 69 | 10.460 | -14.370 | 10.470 |
| 70 | 10.460 | -14.370 | 10.470 |
| 71 | 10.460 | -14.370 | 10.470 |
| 72 | 10.460 | -14.370 | 10.470 |
| 73 | 10.460 | -14.370 | 10.470 |
| 74 | 10.460 | -14.370 | 10.470 |
| 75 | 10.460 | -14.370 | 10.470 |
| 76 | 10.460 | -14.370 | 10.470 |
| 77 | 10.460 | -14.370 | 10.470 |
| 78 | 10.460 | -14.370 | 10.470 |
| 79 | 10.460 | -14.370 | 10.470 |
| 80 | 10.460 | -14.370 | 10.470 |
| 81 | 10.460 | -14.370 | 10.470 |
| 82 | 10.460 | -14.370 | 10.470 |
| 83 | 10.460 | -14.370 | 10.470 |
| 84 | 10.460 | -14.370 | 10.470 |
| 85 | 10.460 | | |

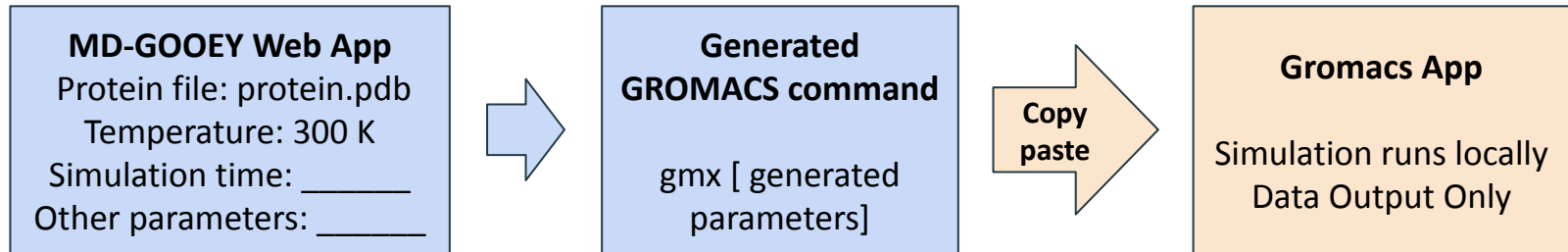
Main Problem

- **Runs Only on the Command Line**
 - Companion Website generates commands with **Limited Functionality**
 - **Applications Disconnected**
- Must be run on users' local computers
 - **Requires Full Download + Setup Per User**
- Desired Computer **Cannot Be Connected to the Internet**



Current state of GROMACS

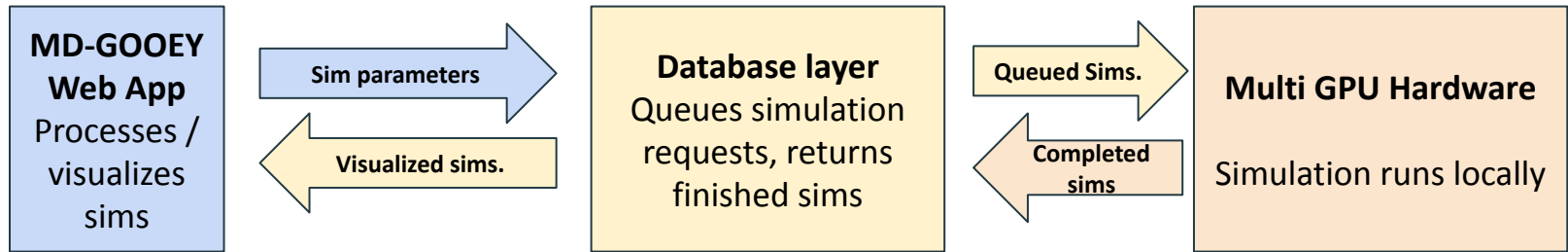
Lacks Full Stack User Flow



Needs:

- **UX Improvements**
- **Accessibility to The Best Hardware**
- **Security**

Solution architecture: MD-Gooley full stack adds a database and hardware layer



- **Database layer receives and queues all simulations** sent from users globally
 - Receives requests from the hardware layer for queued sims
- New stack incorporates hardware with robust capacity to process all simulations
 - Users no **longer need to set up and run GROMACS** locally
- **Front end provides visualized results** of simulations built from text file output

Tech Stack



Frontend

- React, Vite, Typescript - enhance frontend to accommodate functionality

Backend

- Neon for the database
- LMU authentication



Agile Sprint Schedule

Sprints planned out as **2 week blocks**, with **7 sprints planned in total**:

Sprints **#1-3**:

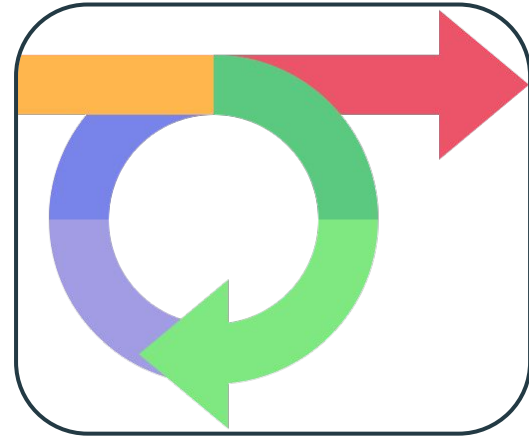
- Base full stack functionality (GUI, database, hardware)

Sprints **#4-5**:

- UX layer / particle visuals completed

Sprints **#6-7**:

- Quality assurance
- Stretch goals (AI chatbot interface, etc.)



Customer Communication:

- Communicate **bi-weekly** minimum
- Via Outlook, Slack, and Zoom

Seeking **consistent feedback / iteration**

Why We Want to do it?

- **Apply our collective skill** set including:
 - Game Design, UI Design, Designing for Customers, Consulting, Prior Involvement with the Project, Graphic Design, API Design and Usage, Quality Assurance Experience, IT and End-User Security
- **Expand our knowledge**
 - Designing for **security, database with queuing capabilities**
 - Potentially: re-formatting data for visualization, graphics visualizations in the browser
 - Potentially: Developing an agent, RAG, training model for a specific subset of tasks
 - Molecular Physics
- Opportunity to **design for a unique subset of users**
 - Physics background
 - Low-no coding ability

Questions?